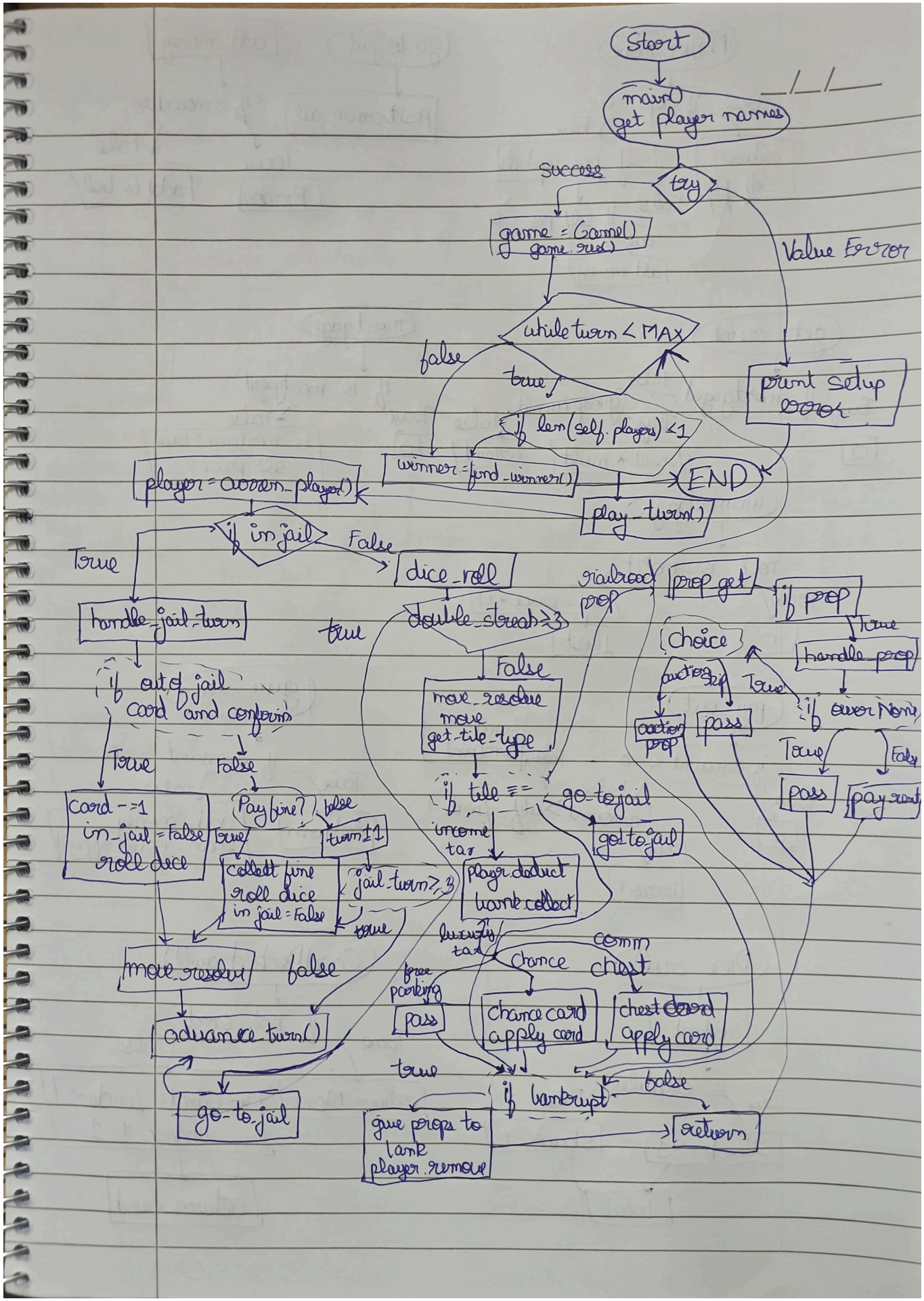
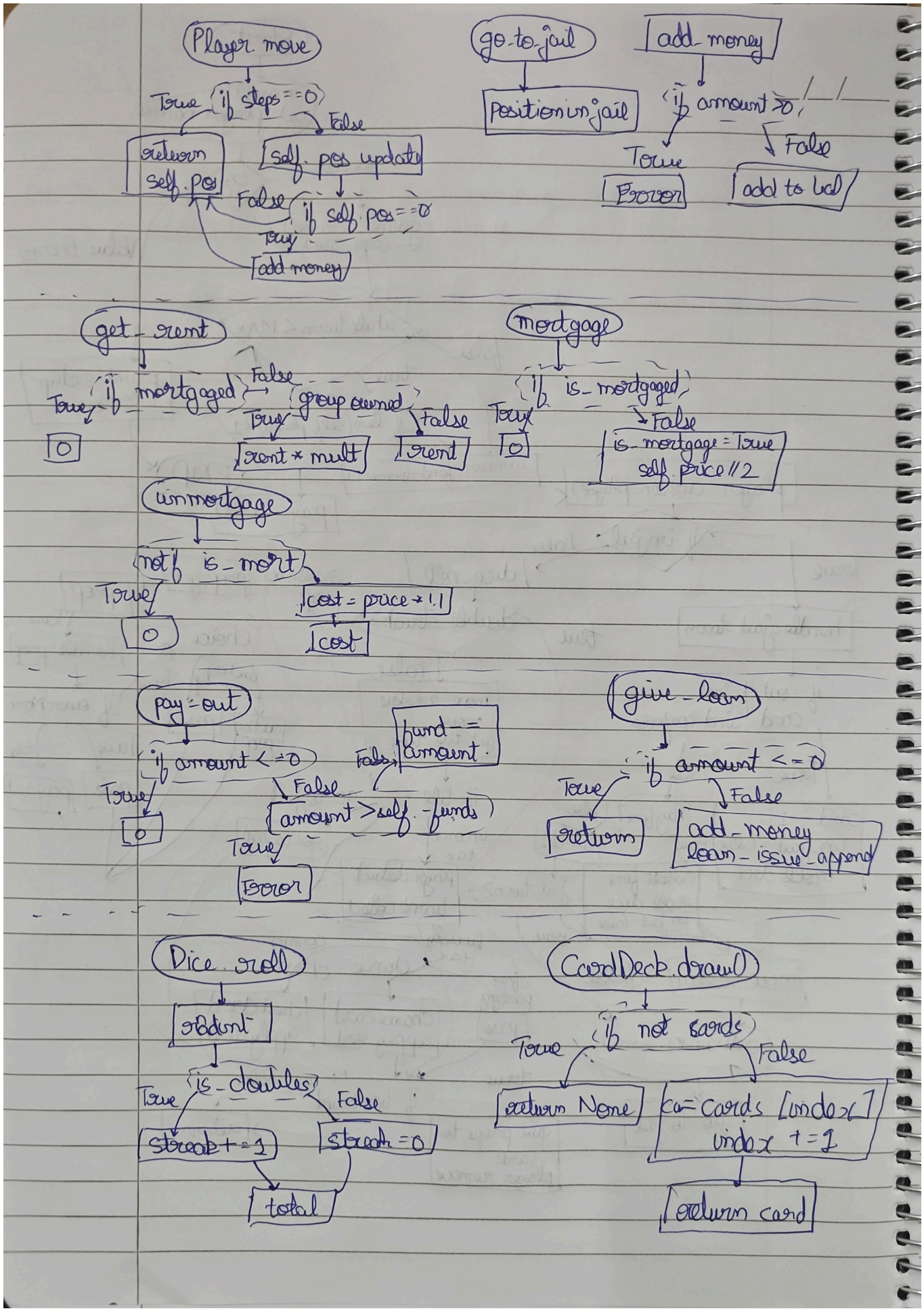
Whitebox testing

1.1 Control flow graph


Start
main()
get player names
try
success
Value Error
game = Game()
game.run()
print setup error
while turn < MAX
false
true
if len(self.players) < 1
winner = find_winner()
END
play_turn()
player = current_player()
if in jail
True
False
handle_jail_turn
if out of jail card and confirm
True
False
card -= 1
in_jail = False
roll dice
Pay fine?
false
true
turn += 1
collect fine
roll dice
in jail = False
jail_turn >= 3
true
move_result
false
advance_turn()
go_to_jail
dice_roll
double_streak >= 3
true
False
move_result
move
get_tile_type
if tile ==
go_to_jail
income tax
go to jail
player.deduct
bank.collect
luxury tax
chance
comm chest
free parking
pass
chance card
apply card
chest card
apply card
if bankrupt
true
false
give props to bank
player.remove
return
railroad
prop
prop.get
if prop
True
handle_prop
Choice
auction
skip
True
if owner None
True
False
auction prop
pass
pass
pay rent


Player. move
if steps == 0
True
False
return self. pos
self. pos update
if self. pos == 0
True
False
add money
go_to_jail
position in jail
add_money
if amount > 0
True
False
Error
add to bal
get_rent
if mortgaged
True
False
0
group owned
True
False
rent * mult
rent
mortgage
if is_mortgaged
True
False
0
is_mortgage = True
self. price // 2
unmortgage
not if is_mort
True
0
cost = price * 1.1
cost
pay_out
if amount <= 0
True
0
False
amount > self. funds
False
fund -= amount.
True
Error
give_loan
if amount <= 0
True
return
False
add_money
loan_issue_append
Dice. roll
roll
is_doubles
True
False
streak += 1
streak = 0
total
CardDeck. draw()
if not cards
True
return None
False
c = cards [index]
index += 1
return card

1.2 Code Quality analysis

Iteration 1

- Added docstrings to functions in main.py

Iteration 2 (property.py)

- Addded module docstring
- Removed 2 attrs from Property class (both were unused)
- Removed argument from Property class
- Removed the else after return in Property.unmortage
- Added docstring to PropertyGroup

Iteration 3 (cards.py)

- Formatted to have smaller length lines
- Added docstring

Iteration 4 (boards.py)

- == True to just checking true
- Added docstring

Iteration 5 (player.py)

- Remove import
- Add final newline
- Added docstring
- Remove unused var
- Removed is_elim attr

Iteration 6 (game.py)

- Added docstring
- Made fstring to normal string
- Removed unused imports
- Removed unnecessary parens after not keyword
- Added final newline
- Removed setting player is eliminated
- Consolidated chance_deck and community_deck into a dictionary to reduce instance attributes

- Updated all references to use new decks dict structure
- Removed running attribute
- Extracted card action handling into separate private methods to reduce branching error

Iteration 7 (config.py)

- Added module docstring for documentation

Iteration 8 (player.py)

- Removed unused variable from move() method

Iteration 9 (dice.py)

- Added module docstring
- Removed unused import of BOARD_SIZE
- Moved doubles_streak attribute initialization into **init**

Iteration 10 (bank.py)

- Added module docstring
- Added Bank class docstring for clarity
- Removed unused math import

Iteration 11 (ui.py)

- Added module docstring
- Specified exception type (ValueError)

1.3 White Box Test Cases

Bugs

1. PropertyGroup.all_owned_by() Uses `any()` Instead of `all()`: The method uses `any()` which returns True if **any** property in the group is owned by the player, instead of checking if **all** properties are owned by the player.
2. Bank.collect() Docstring Doesn't Match Implementation: The docstring claims negative amounts are "silently ignored," but the code actually subtracts them from the bank's funds and records them in totals.
3. When giving move() function 0 steps, it wasn't checking the steps
4. Winner selection should use max networth.
5. Allow property purchase with exact balance, just not less than that

6. Award Go salary when passing the Go area not just when landing on it
7. Give rent payments to the property owner after deducting it from the spot lander
8. Deduct balance when paying the jail fine
9. After property selling transfer money to the property seller
10. Changed dice range to 6
11. Calculate networth using property and balance

Test Case Details and Explanations

Player Tests

- Tests moving within board bounds and verifies position updates correctly
- Tests landing exactly on position 0 and verifies GO salary ($200) is collected
- Tests moving from position 38 by 5 (wraps to position 3)
- Buy property checks if property is added to list
- Go to Jail function works or not
- Try to remove non existent property, it should throw an error
- Test if transactions are working correctly

Property Tests

- Tests when player owns all properties in group and verifies rent doubling condition works
- Tests when player owns only some properties (1 of 2)
- Test mortgage value is half of price
- is_mortgaged state works properly
- Rent function for owned/unowned/mortgaged property works correctly

Bank Fund Management Tests

- Tests normal fund collection
- Tests negative amount collection
- Tax collection, fund payouts, loans all work correctly
- Test normal loan amount, negative amount as well
- Test if paying funds are respected by the amount in bank

Property group tests

- Validates the grouping of properties by color and tracks how many properties in a specific group are owned by each player.

- Specifically tests the logic for determining if a single player owns every property in a color group.

TestCardDeck

- Ensures that drawing cards follows the established order and that the deck "wraps around" back to the first card once the last card is drawn.
- Tests "peeking" at the top card without removing it and counting remaining cards in the deck.

TestGame & TestGameIntegration

- Verifies the game cycles through players in the correct order.
- Tests property purchases and trades between players, ensuring funds and ownership are transferred only if the buyer has sufficient cash.
- Confirms that drawing cards triggers the correct game effects, such as collecting money, paying fees, or going to jail.
- Validates that when a player goes bankrupt, they are removed from the game and their properties are returned to the bank (owner set to None).
- Tests integration scenarios, such as being sent to jail immediately after rolling three consecutive doubles.

TestBoard

- Confirms that specific positions on the board correspond to the correct tile types (e.g., Position 0 is "Go", Position 10 is "Jail").
- Verifies that the board can return the correct Property object based on a numerical position index.